Yash Mehta

Data Engineer | Python Developer | Data Analytics & Machine Learning

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PROFESSIONAL SUMMARY

Data-focused Computer Science professional with hands-on experience in Python-based data processing, database integration, predictive analytics, and data-driven application development. Proficient in Pandas, NumPy, SQL, MySQL, MongoDB, and time-series forecasting techniques. Experienced in developing forecasting models, analytical dashboards, and machine learning pipelines involving structured and unstructured data. Developed multiple end-to-end projects and research applications, including published work in machine learning and deep learning. Interested in Data Engineering, Analytics Engineering, and data-centric software development.

TECHNICAL SKILLS

- **Programming**: Python, Java, C++, JavaScript
- **Data Processing:** Pandas, NumPy, Data Cleaning, Data Transformation, Feature Engineering
- **Databases***:* MySQL, MongoDB, SQL
- **Data Analytics**: Exploratory Data Analysis, Data Visualization, Statistical Analysis, Predictive Analytics
- **Forecasting:** ARIMA, Prophet, LSTM, GRU, Transformers
- **Machine Learning:** Scikit-learn, Random Forest, Model Evaluation
- **Visualization:** Chart.js
- **Backend:** Flask, REST APIs
- **Tools:** Jupyter Notebook, Google Colab, Git, Apache

SELECTED DATA & ANALYTICS PROJECTS

Amazon Stock Price Prediction | Financial Time-Series Analytics | 2025

- Developed Python-based forecasting workflows for financial time-series prediction.
- Implemented and evaluated ARIMA, LSTM, GRU, and Transformer-based approaches.
- Achieved an RMSE of 0.03285 in forecasting experiments.
- Designed a custom TimeSeriesTransformer supporting multivariate inputs.

Retail Analytics Dashboard - Vishleshan | Data Visualization | 2023

- Developed a Flask and MySQL-based retail analytics dashboard.
- Implemented data retrieval and analytical workflows for sales monitoring.

- Integrated Chart.js to visualize sales trends and business insights.
- Presented the application to 200+ industry experts at Bennett University.

Fake News Detection | Text Analytics & Machine Learning | 2025

- Developed a text classification pipeline using TF-IDF and Random Forest.
- Integrated sentiment analysis into the classification workflow.
- Achieved 87% validation accuracy.
- Developed a Flask application for interactive model demonstration.
- Presented the research at SocProS-25, IIT Roorkee.

Fog Detection Using Hybrid Deep Learning Models | 2025

- Worked with the FRIDA2 dataset for fog detection research.
- Developed a hybrid CNN and Vision Transformer-based approach.
- Reported a 15% improvement in accuracy on real-world haze datasets.
- Integrated OpenCV for image processing.

Room Allocation Tool | Database Management | 2023

- Developed a Java and MySQL-based room allocation application.
- Implemented automated allocation and database synchronization.
- Supported workflows involving 100+ students.
- Reduced manual allocation effort by approximately 95%.

PROFESSIONAL EXPERIENCE

Software Intern | MNNIT Allahabad | 2024

- Developed Flask REST APIs for academic service applications.
- Improved backend throughput by 20%.
- Worked with backend workflows and application integration.
- Collaborated through Agile development processes.

RESEARCH EXPERIENCE

Blockchain in Autonomous Vehicles | Bennett University | 2024

- Simulated blockchain protocols for autonomous vehicle communication.
- Evaluated spoofing-related security risks in test environments.
- Reported a 35% reduction in spoofing risks under experimental conditions.

RESEARCH & PUBLICATIONS

1) Fake News Detection Using Machine Learning and Sentiment Analysis Integrated in a Flask Web Application.

Springer Nature, Real-World Applications of Machine Learning, 2026.

DOI: https://doi.org/10.1007/978-981-95-7134-5_23

Earlier Conference Contribution: SocProS-25, Soft Computing for Problem Solving, IIT Roorkee, 2025. Book of Abstracts, page 90.

2) Fog Detection Using Hybrid Deep Learning Models: A Multi-Modal CNN-LSTM Approach.

IEEE ICCCA, 2025.

DOI:10.1109/ICCCA66364.2025.11325690

CERTIFICATIONS

- NVIDIA - Fundamentals of Deep Learning
- IBM - IBM Machine Learning Professional Certificate
- Google Cloud - Introduction to Generative AI
- IBM - Generative AI Fundamentals
- IBM - AI Foundations for Business
- Google - Google Data Analytics Professional Certificate
- IIT Delhi - Quantum Computing & QKD

EDUCATION

B.Tech. - Computer Science & Engineering

Bennett University | 2022–2026 | CGPA: 8.26

Class XII - CBSE | 2021 | 92.4%

Class X - CBSE | 2019 | 85.8%

Maharshi Patanjali Vidya Mandir